Map Compressor

Turn large drone maps into small, see-through overlays for TAK

User Guide · Version 0.1.1 · November 2026

What it does

Drone mapping software (PIX4Dreact, PIX4Dengine, DroneDeploy, WebODM and others) produces orthomosaic maps that are often **hundreds of megabytes**. Those files are slow to send, too big for many phones and tablets, and the empty area around the flight shows up as a **black box** in TAK.

Map Compressor shrinks the map so it sends quickly and loads on any TAK device. It keeps the map in the right place, and it keeps the empty border **see-through** so only your imagery shows on the map.

223 MB → ~31 MB

A real fire-scene drone map, converted on a phone in about a minute

See-through

No black box around the imagery. The basemap shows around it.

100% on-device

No account, no cloud, no upload. Safe for sensitive imagery.

Three ways to use it

📱 ATAK plugin

Convert on an Android phone or tablet, inside ATAK. The map is added straight away and you can share it from ATAK.

💻 Laptop page

One file you double-click on any Windows, Mac or Linux computer. No install and no internet needed.

📡 Any TAK client

The finished file is made for ATAK, iTAK, TAK Aware, WebTAK and CloudTAK.

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Downloads

Click the link for the version you need. The download starts straight away; no account or sign-up is needed.

📱 **ATAK 5.8**	Download Map Compressor for ATAK 5.8 `github.com/AM-Ripple2/map-compressor/releases/download/v0.1.1/ATAK-Plugin-MapCompressor-0.1.1--5.8.0-civ-release.apk`
📱 **ATAK 5.7**	Download Map Compressor for ATAK 5.7 `github.com/AM-Ripple2/map-compressor/releases/download/v0.1.1/ATAK-Plugin-MapCompressor-0.1.1--5.7.0-civ-release.apk`
📱 **ATAK 5.6**	Download Map Compressor for ATAK 5.6 `github.com/AM-Ripple2/map-compressor/releases/download/v0.1.1/ATAK-Plugin-MapCompressor-0.1.1--5.6.0-civ-release.apk`
💻 **Laptop / desktop**	Download the Map Compressor page (works offline) `github.com/AM-Ripple2/map-compressor/releases/download/v0.1.1/map-compressor-offline.html`

Which ATAK download do I need?

Open ATAK's **About** screen and look at the version number. Only the first two numbers matter:

ATAK shows…	Download
5.8.x.x	ATAK 5.8
5.7.x.x	ATAK 5.7
5.6.x.x (for example 5.6.0.13)	ATAK 5.6

✅ **Signed by TAK.gov**

All three ATAK downloads went through the TAK.gov third-party plugin pipeline. They load in the normal ATAK-CIV from the Play Store or TAK.gov. You don't need a developer version of ATAK.

Downloading on a computer instead of the device?

Save the file, then copy it to the phone or tablet with a USB cable, email, or a shared drive. Put it somewhere easy to find, such as the **Download** folder.

All downloads in one place

To see every file, or check for a newer version later, open github.com/AM-Ripple2/map-compressor/releases/latest and click a file name under **Assets**.

📱 Using the ATAK plugin

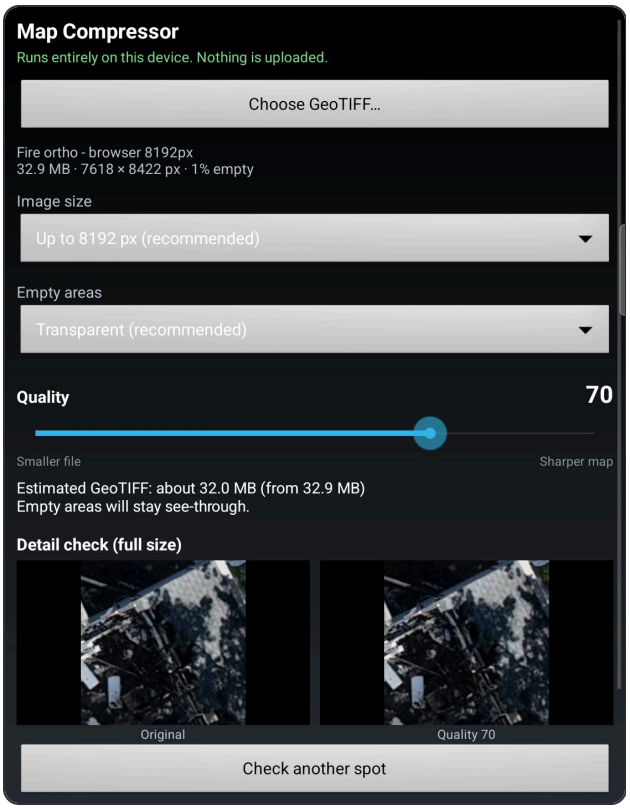
Install (once)

1. Download the file for your ATAK version (see page 2) on the phone or tablet.
2. Tap the downloaded file, then tap **Install**. If Android asks, allow installs from your browser or file manager.
3. Open ATAK and go to **Settings** → **Tool Preferences** → **Plugins**. Find **Map Compressor** and load it.
4. ATAK may show that the plugin was signed through the TAK third-party pipeline. That's expected; accept it.

Updating later: install the new file over the old one. You don't need to uninstall first.

Convert a drone map

1. Put the drone map file (`.tif` or `.tiff`) on the device, for example in the **Download** folder.
2. In ATAK, open **Map Compressor** from the toolbar or the **Tools** menu. A panel opens on the side (right).
3. Tap **Choose GeoTIFF...** and pick your file. Below the button, the panel shows the map's name, file size, pixel size and how much of it is empty.
4. Leave the settings on the recommended defaults, or adjust them (see page 6):
   - **Image size:** Up to 8192 px (recommended)
   - **Empty areas:** Transparent (recommended)
   - **Quality:** 70. Drag toward *Smaller file* or *Sharper map*. The **Estimated GeoTIFF** size updates as you drag.
5. *Optional:* the **Detail check** shows the same spot from the *Original* and the compressed version side by side, at full size. Tap **Check another spot** to look somewhere else.



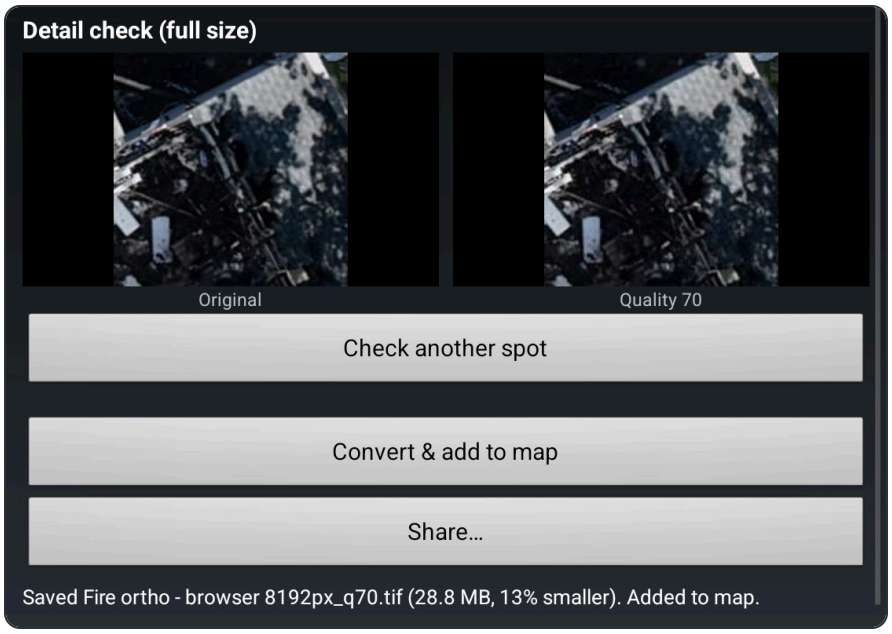
The Map Compressor panel with a map loaded

📱 Using the ATAK plugin (continued)

Convert a drone map

6. Scroll to the bottom of the panel and tap **Convert & add to map**. A progress bar shows how it's going; a large drone map takes about a minute.

7. When it's done, the overlay is added to the map and the map zooms to it. The last line of the panel shows the saved file name, its size and how much smaller it is.



Bottom of the panel after converting. This example input was already compressed, so it only shrank 13%; the original 223 MB drone map shrank by 86%.

Where is the file saved?

In the device's `atak/grg` folder, named after the map and the quality, for example `FireScene_q70.tif`. ATAK lists it with your other overlays.

Share it with your team

Tap **Share...** after converting. ATAK's normal send screen opens, so you can send the map to contacts, a TAK Server or a Data Sync mission. Recipients get the usual "file received" message.

💻 Using the laptop page

A single web page that runs on any computer: Windows, Mac or Linux, in Chrome, Edge, Firefox or Safari. It doesn't need installing, admin rights or internet. It also works from a USB stick or shared drive.

1. Download **map-compressor-offline.html** (see page 2) and double-click it. It opens in your web browser.
2. Check the green banner at the top left: **"Offline: nothing leaves this device."**
3. Click **Choose a GeoTIFF map**, or drag your drone map file onto that box.
4. Set **Image size**, **Fill for empty areas** and **Quality**. The defaults are a good start (see page 6). The file size estimate updates as you change them.
5. Check the preview on the map. It shows exactly where the overlay will appear in TAK. Use **Preview opacity** to see through it.
6. *Optional:* in the **Detail check**, click a spot on the small map to compare the original and compressed image at full size.
7. Click a download button:
   - **Download GeoTIFF**: **recommended**. One file, with a see-through border.
   - **Download KMZ**: only if your app needs KMZ. The border is filled black or white.
   - **Download Data Package**: the GeoTIFF packed for TAK Server or Data Sync.
8. Send or copy the file to your TAK device (see page 7).

Optional satellite background

Ticking **Show online satellite basemap (Esri)** shows satellite imagery behind your map, so you can check it lines up. This needs internet and downloads map tiles from Esri, which shows Esri the *area* you're looking at (never your drone image). The banner turns orange while it's on. It's always off when you open the page.

Grey map before loading a file?

That's normal. With the satellite background off, the map is plain grey until you load a drone map.

Technical users: a command-line version for converting whole folders at once is also available. See the user manual at github.com/AM-Ripple2/map-compressor.

Settings explained

Image size

Limits how many pixels wide or tall the finished map can be. Smaller means a smaller file.

Option	When to use it
Up to 8192 px (recommended)	The best balance for most drone maps: sharp at normal zoom, and a much smaller file.
Full resolution	When you need every detail, such as reading small text or inspecting damage closely. The file is much bigger.
Up to 4096 px	Slow networks, many recipients, or older devices. Fine for an overview.
Up to 2048 px	The smallest file, for a quick overview or a very slow link.

Empty areas

A drone map is an irregular shape inside a rectangle, and the area outside the flight has no imagery.

- **Transparent** (recommended): the empty area is see-through, so the TAK basemap shows around your imagery.
- **Black** or **White**: fills the empty area with a solid colour. Use it only if an app has trouble showing the map.

Quality

A slider from **Smaller file** to **Sharper map** (10 to 95, default **70**).

- **60–75** looks good for most drone maps.
- Lower values give a smaller file, but fine detail gets soft or blocky.
- Above about 85, the file grows a lot for little visible improvement.

To judge it, use the **Detail check** on a spot with fine detail, such as text, vehicles or road lines, and compare *Original* with *Compressed*.

Quick picks

Situation	Image size	Quality	Download as
Everyday use	Up to 8192 px	70	GeoTIFF
Slow network or many recipients	Up to 4096 px	60	GeoTIFF
Close inspection (text, damage)	Full resolution	80	GeoTIFF (check the size before sending)
App can't show see-through maps	Up to 8192 px	70	KMZ

Results depend on the image: busy scenes such as forest or rubble don't shrink as much as open ground or rooftops.

Opening the map in each TAK app

App	How to open the map	Status
ATAK	**Import Manager** → **Image Overlay File** → pick the `.tif` or `.kmz`. Data Packages open from Import Manager too. The ATAK plugin adds the map for you.	✅ Tested on 5.6 5.7 / 5.8 expected to match
iTAK	Import the GeoTIFF or Data Package (Files app, AirDrop, or received from another user).	Please report results
TAK Aware	Import the GeoTIFF or Data Package.	Please report results
WebTAK / CloudTAK	Upload the GeoTIFF or Data Package through the import or files menu.	Please report results

To send the map to a team, share the GeoTIFF or Data Package the way you normally share files: ATAK's send screen, a TAK Server Data Package, or a Data Sync mission.

Troubleshooting

Problem	What to try
Map Compressor isn't in ATAK's plugin list	Check you downloaded the file for your ATAK version (page 2). Restart ATAK and look in Settings → Tool Preferences → Plugins.
ATAK says the signature is invalid	An old test version may be installed. Uninstall Map Compressor, then install the download from page 2.
Can't find my file when choosing it	The file must end in `.tif` or `.tiff` and be saved on the device (for example in Download), not only in a cloud app.
The map is in the wrong place	The file may be missing its location data. Please report it, with the software that made the map.
Black box around the imagery	Download as **GeoTIFF** with Empty areas set to **Transparent**. KMZ can't be see-through.
Blurry when zoomed in	Raise **Quality**, or choose a bigger **Image size**.
File still too big	Lower **Image size** first (8192 → 4096), then lower **Quality**.
Very slow, or the device runs out of memory	Very large maps (over 500 MB) need a lot of memory. Use a smaller Image size, or convert on a laptop.

Questions, problems or feedback?

Reply to the email this guide came with, or report it at github.com/AM-Ripple2/map-compressor/issues. It helps to include your device, which TAK app and version you use, the map file's size, and the software that made it.